Delft University of Technology

Faculty of Electrical Engineering, Mathematics and Computer Science

ET4351 Digital VLSI Systems on Chip

Project Report — Academic Year 2025/2026

Group 9

Team Members & Individual Contributions

#	First Name	Last Name	Student ID	Individual Contribution
1	Franz-Josef	Zuaiter	5725992	Final EE Design + Design and implementation of Radix 4-4-2 FFT (EE)
2	Romeu	Longo Malinski	6327087	Final EE Design + Design and implementation of low-activity twiddle-handling optimizations (EE)
3	Shangong	Lin	6408931	Microarchitecture design and implementation for High Performance (HP) target.
4	Jiahui	Que	6551998	Final EE Design + Design and implementation of RFFT Radix 4 FFT (EE) + Post-Layout Simulation (EE) + Validation of the design (EE)
5	Anastasis	Kazakos	6398308	Comparison of the PnR results/reporting PnR and floorplaning (HP)
6	Alieldin	Sakr	6416705	Final EE Design+ Design and implementation of RFFT with Radix 4 + Final EE Hold violations solving (EE)
7	Yaonan	Hu	6573657	Waveform simulation + Documentation (HP)
8	Daniel	Tyukov	5714699	Final EE Design + Design and implementation of low-activity twiddle-handling optimizations (EE) + registers implementation and manual clock gating
9	Alessandro	Cagnacci	6524842	PNR scripts (HP)
10	Leonardo	Castello	6553583	Post-layout simulation, VCD and final sign-off (HP)

"Primary Responsibility": briefly state each member's main contribution (e.g., "Butterfly unit RTL", "Genus synthesis & clock gating", "Post-layout sim & VCD").

Report Guidelines & Knock-Out Criteria

1) **Format:** IEEE double-column, compiled to PDF. Font size: 10 pt (IEEE default). Use this template.
2) **Page limit:** The main body (Sections I–IV) must fit within **6 pages** (≈3 pages per design). This cover page and all appendices are **excluded** from the page count.
3) **Structure:** You **must** follow the mandatory section structure defined in this template. Do not add, remove, or rename top-level sections (I–IV).
4) **Two designs:** Present both a **High-Performance (HP)** and an **Energy-Efficient (EE)** accelerator. Shared aspects should be described once; per-design subsections are provided. **Targets at 12 MHz:** HP — latency $< 61.00\,\mu$s (baseline); EE — accelerator power < 0.403 mW (baseline). Clock period is **free to change**.
5) **Submission checklist** (all mandatory):
   - ❑ Report (PDF) with cover page filled in
   - ❑ SystemVerilog/Verilog source files (cleaned & commented)
   - ❑ All synthesis and P&R scripts
   - ❑ `finaldesign_hp/` and `finaldesign_ee/` directories (each with `accel_audio.hex`, `et4351.phys.sdf`, `et4351.phys.v`)
   - ❑ All signoff reports: included as files in the project zip **and** as screenshots/excerpts in the PDF appendix
   - ❑ Activity annotation and full power reports: included as files in the project zip **and** as screenshots/excerpts in the PDF appendix
6) **Signoff requirements** (per design): core area 596.4 µm × 596.4 µm; setup & hold timing clean, fully DRV clean (`max_tran` violations are acceptable but risky near zero slack); connectivity, geometry, antenna reports clean; functionally correct FFT on variable-length audio input.
7) **Quality:** Proper use of technical terms. Reasonably proofread. Clean, readable diagrams.
8) **Deadline:** Friday, April 10, 2026 at 16:59 CET.

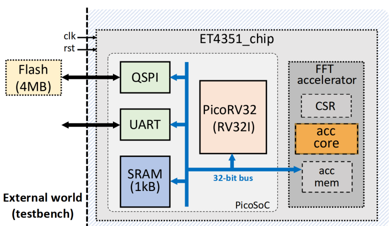
ET4351 Project Report — Group 9

I. High-Performance (HP) Accelerator

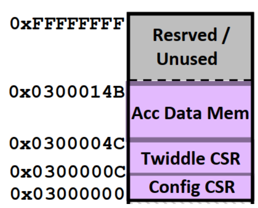
A. Architecture and Design Methodology

The High-Performance (HP) accelerator targets a per-chunk FFT latency below the 61.00 μs baseline for N=32 complex samples, with no constraint on clock frequency. The design implements a decimation-in-time (DIT) radix-2 FFT across five butterfly stages, each processing 16 operations. The final architecture achieves 121 cycles at 66.2 MHz, yielding a latency of $\sim$1.83 μs—a $\sim$33$\times$ improvement over the baseline.

SoC Integration: The accelerator is integrated into a PicoSoC built around the PicoRV32 (RV32I) CPU core, connected alongside a QSPI flash controller, UART, and GPIO via a shared 32-bit memory-mapped `iomem` bus (Fig. 1a). The CPU orchestrates the FFT by writing audio samples to the accelerator's data memory and loading pre-computed twiddle factors—stored in flash by the offline build toolchain—into the accelerator's CSR registers, then asserting an enable bit and polling a done flag.



(a) SoC block diagram showing the PicoRV32 core and the FFT accelerator connected via the 32-bit `iomem` bus.



(b) Memory map (partial) of the HP design, showing the accelerator's CSR and data memory regions.

Fig. 1: HP SoC architecture and memory map.

Bottleneck Analysis and HW/SW Co-Design: The baseline accelerator executes the 32-point DIT radix-2 FFT through a 13-state interleaved FSM that issues individual SRAM reads and writes for every operand, requiring 732 clock cycles per chunk at 12 MHz (61.00 μs). Profiling reveals that memory transfers account for roughly 88% of total cycles, while butterfly arithmetic contributes only a small fraction. The HP design therefore targets both *memory bandwidth* and *compute path length*: architectural changes reduce the cycle count from 732 to 121 (6$\times$), while pipeline-enabled frequency scaling raises the clock to 66.2 MHz (5.5$\times$), combining for a $\sim$33$\times$ latency reduction.

Accelerator Architecture: The accelerator core (Fig. 2) combines five architectural features into a coherent datapath, each motivated by a specific bottleneck exposed by its predecessor. First, a **register-file datapath** holds all 32 complex samples (64 words $\times$ 24-bit) in flip-flops, confining accelerator memory access to bulk LOAD and STORE phases and eliminating the per-butterfly SRAM reads/writes that dominate the baseline cycle budget. Second, **software twiddle preload via CSR registers** removes twiddle factor accesses from the timed computation window: the firmware pre-computes all 16 unique W_{32}^{k} pairs as Q12 fixed-point values and writes them to 16 packed 32-bit CSR words ({`tw_im[15:0]`, `tw_re[15:0]`}) before asserting enable, making every twiddle simultaneously available by index. This is a prerequisite for the third feature: **two parallel butterfly units** that process two independent operations per cycle, halving the issue count per FFT stage from 16 to 8. Fourth, a **4-stage micro-pipeline** (Fig. 3) splits the butterfly's combinational multiply-add chain into FETCH $\rightarrow$ MUL1 $\rightarrow$ MUL2 $\rightarrow$ ADD stages, each containing roughly one quarter of the critical path delay. A new butterfly pair enters every clock cycle (unit throughput), and each FFT stage completes in 8 fetch + 3 drain = 11 cycles; across five stages this yields 55 compute cycles. The pipeline also breaks the $\sim$19.6 ns combinational path measured at the baseline frequency into four shorter segments, enabling synthesis at 66.2 MHz (15.1 ns period) with timing closure verified post-route. Fifth, a **wide paired memory port with 24-bit datapath** (48-bit, i.e. 2 $\times$ 24-bit) reads or writes one complete complex pair (real + imaginary) per cycle during LOAD and STORE, halving the per-phase transfer time from 64 words $\times$ 1 word/cycle to 32 pairs $\times$ 1 pair/cycle. The internal data width is narrowed from 32 to 24 bits, shrinking register-file storage, memory cell area, and multiplier size (24$\times$16 instead of 32$\times$16) while retaining sufficient dynamic range ($\pm$8 M) for the audio signal path.

The resulting cycle breakdown is: INIT (1) + LOAD (32) + COMPUTE (55) + STORE (32) + FINISH (1) = **121 cycles**. The FSM is reduced from 13 baseline states to five (`INIT`$\rightarrow$`LOAD`$\rightarrow$`COMPUTE`$\rightarrow$`STORE`$\rightarrow$`FINISH`).

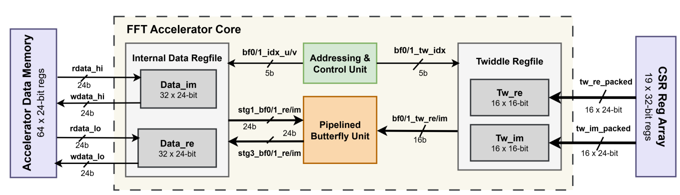


Fig. 2: FFT accelerator core architecture, showing the register file, 2$\times$ pipelined butterfly unit, twiddle CSR interface, and wide paired memory port.

Memory Hierarchy and IP Selection: The accelerator interface preserves the baseline memory map, making the modified core a drop-in replacement (Fig. 1b). Beyond the three baseline configuration CSRs, 16 additional 32-bit registers hold the 16 complex twiddle factors, each packing one real/imaginary pair into a single word, which keeps every twiddle available to

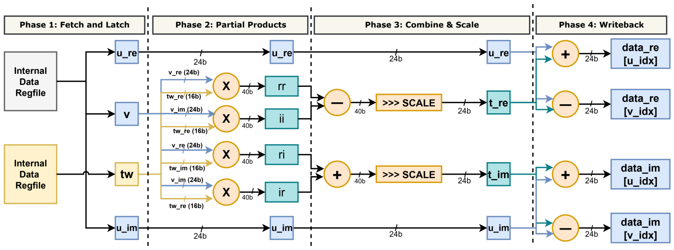


Fig. 3: Single butterfly lane of the 4-stage micro-pipeline. Two such lanes operate in parallel, each processing one butterfly per cycle.

the FFT core at zero cycle cost during the timed computation window.

The accelerator data memory (`accelerator_mem`) is implemented as a synthesised register file rather than an additional SRAM macro. The existing SoC already contains four SRAM1RW256×8 instances from the `saed32sram` library, serving as PicoRV32 working memory and occupying 34,535 μm^2 in total ($\approx$8,634 μm^2 each). Reusing this macro for the accelerator's data store is infeasible: its single read-/write port cannot supply the simultaneous narrow (32-bit, CPU-side) and wide (48-bit, FFT-core-side) interfaces required by the HP architecture, making it inefficient for accelerated computing. A synthesised register file of 64 words $\times$ 24-bit occupies 31,330 μm^2—roughly the same area as the total SRAM macro—but provides true dual-port access that enables the wide paired memory port: one complete complex pair ($2 \times$ 24-bit) is read or written per cycle during LOAD/STORE, halving the per-phase transfer time from 64 to 32 cycles and saving 64 cycles per FFT chunk. This area overhead is justified by the cycle-count reduction it helps deliver. The design exploits the interleaved real/imaginary memory layout (`[re[0], im[0], re[1], im[1], ...]`): a 5-bit pair address indexes one of 32 complex pairs, and two 32:1 mux trees deliver both words of the selected pair in a single cycle.

B. Implementation

SDC Constraints and Rationale: The timing constraints were adapted from the baseline SDC to target the HP operating point of 66.2 MHz (15.1 ns clock period), established by the architectural design. The clock period was reduced from the baseline 83.33 ns to 15.1 ns, reflecting the 5.5× frequency scaling achieved by the 4-stage pipeline (described in Section I-A).

The global clock uncertainty was reduced from 0.25 ns to 0.15 ns across all clocks. The original 0.25 ns value was overly pessimistic relative to the actual post-CTS skew. Particularly on `flash_clk`, where measured skew was approximately 0.025 ns, causing artificially inflated hold requirements and contributing to the hold violations observed in post-route. Reducing the uncertainty to 0.15 ns, brought the constraint closer to the physical reality of the clock tree while the safe margin was retained. All other constraints, such as `INVX1` drive strength on inputs, 0.05 pF output load, QSPI I/O delays, and false paths on reset and UART, remained unchanged from the baseline, as these reflect the fixed SoC boundary conditions.

Place-and-Route Tool Settings: In `5_place.tcl`, three targeted changes were made relative to the baseline script. Timing effort was raised from `medium` to `high` in `setPlaceMode`, directing Innovus to spend more optimisation effort, resolving critical paths during global placement. That was considered important given the 15.1 ns target period. Power-driven placement was disabled (`-place_global_activity_power_driven false`), removing the baseline's power-optimised cell spreading which conflicted with timing closure at 70.7% utilisation. Two additional optimisation directives were added: `setOptMode -effort high` and `setOptMode -setupTargetSlack 0.050`, the latter targeting a +50 ps setup margin to provide headroom against post-route degradation.

In `7_route.tcl`, the routing configuration itself was carried over from the baseline. Our addition was a dedicated hold fixing block after the standard post-route optimisation pass. This block designates small buffer cells (`BUFHD1`, `BUFX1`, `BUFX2`, `BUFX4`) as hold-fixing candidates, permits cell resizing and overlap during hold repair (`-fixHoldAllowResizing true`, `-fixHoldAllowOverlap true`), sets a hold target slack of +20 ps (`-holdTargetSlack 0.020`), and sets the violation fixing threshold to 0 ns to ensure no hold violation goes unaddressed. A separate `optDesign -postRoute -drv` pass follows to recover any slew violations introduced by the hold buffering. This strategy, setup optimisation first, then targeted hold and DRV recovery, was necessary to fix the hold and DRV violations present after the initial route.

Signoff Verification: Timing was checked at every major PnR stage via `timeDesign` for both setup (max) and hold (min) corners, with reports saved incrementally after placement, post-route, and post-hold-optimisation. Static power reports at the typical corner were generated at each stage to track power evolution through the flow.

Floorplan & Congestion: The floorplan uses the fixed 596.4 µm × 596.4 µm core area, reaching 70.7% utilisation (251,293.9 µm²). This density stems from the architectural choice of a synthesised register file for the accelerator memory, which requires more standard cells than an SRAM macro, described in the "`Architecture and Design Methodology`". The floorplan structure itself was kept consistent with the baseline. To manage congestion at this utilisation level, power driven placement was disabled in favour of timing driven cell spreading. Placement timing and solver effort were both set to high, and a +50 ps setup slack target was added to push the placer toward timing safe positions before routing, reducing the risk of hold violations emerging post-route.

C. Performance Metrics and Validation

This section describes the methodologies used to validate the physical design and extract the final Power, Performance, and Area (PPA) metrics.

Design Validation and Post-Layout Simulation: After completing the synthesis and place-and-route workflow, the design was checked for physical and electrical integrity. Final reports were checked to verify that the layout was free of DRC violations and met all timing requirements for both setup and hold constraints. Also, Design Rule Violations (DRV) were verified: no DRV errors were found except for *max_tran* violations listed in Table I.

Subsequently, post-layout simulations were conducted for both setup and hold timing. The setup simulation was performed using max corners to ensure that data arrives at the capturing flip-flop with sufficient margin before the next clock edge under the slowest possible conditions; on the other hand, the hold simulation was performed using min corners in order to verify that, even under the fastest possible conditions, the signal remains stable for long enough after the clock edge to prevent data corruption.

Test Specifications: To ensure functional integrity after physical implementation, the design was subjected to a rigorous validation process. A checklist of critical elements was verified via post-layout simulation. A detailed summary of these test specifications and the corresponding results is provided in Table IV (see Appendix A.10).

Metric Extraction: Once the design was validated, PPA metrics were extracted for evaluation. Performance was measured by determining the latency of the accelerator for the first data chunk during behavioral simulation. By dividing the execution time by the *15.1 ns* clock period, the number of cycles required for the operation was calculated.

Area metrics were obtained directly from the final post-PNR area report (*report_area.rpt*), providing the silicon footprint of the layout.

The final stage of metric extraction involved post-layout power estimation using activity annotation. This methodology uses a Value Change Dump (VCD) file to capture the switching activity during execution of the accelerator. To ensure the results represent the actual workload, activity was only recorded for the specific time window in which the accelerator was active. This time window was identified once again using the accelerator start time and latency obtained from the behavioral simulation. This process was managed by the script *11.finalPowerReportsicl*, which reads the switching activity extracted during the post-layout simulation and performs power analysis at the typical delay corner. The analysis achieved 100% annotation coverage, and the final results were captured inside the final report *report_power_postRouteVCD.rpt*.

The post-layout results in Table I confirm successful implementation. Timing is clean for both setup (+0.226 ns) and hold (+0.015 ns), with zero DRC violations. The 28 *max_tran* DRV violations are non-critical. The design achieves 66.2 MHz—a 33.4$\times$ speedup over the 61 μs baseline—with the core area of 251,293.9 μm^2 well within the 355,693 μm^2 constraint.

TABLE I: Post-Layout Results

Metric	**Value**
Timing & Area	
Frequency	66.2 MHz
Core Area	251293.9 μm^2
Latency (cycles)	121
Latency (μs)	1.8271 μs
Sign-off	
Setup WNS	+0.226 ns
Hold WNS	+0.015 ns
DRC Violations	0
Critical DRV Violations	0
max_tran violations	28
Power (activity-annotated)	
Accel. Dynamic Power	1.652 mW
Accel. Leakage Power	0.01667 mW
Total Power (SoC)	2.795 mW
Energy ($P \times T$)	5.107 nJ
Annotation Coverage	100 %

Total SoC power increased from 0.403 mW to 2.795 mW (accelerator dynamic: 0.093→1.652 mW; accelerator leakage: 0.013→0.017 mW), yet total energy *decreased* from 24.6 nJ to 5.107 nJ—a 4.8$\times$ reduction—confirming that the latency improvement more than compensates for the higher power draw.aw.onsumption, resulting in a faster but also more energy-efficient design.

II. Energy-Efficient (EE) Accelerator

A. Architecture and Design Methodology

The Energy-Efficient (EE) accelerator targets a reduction in total energy consumption below the 24.6 nJ baseline while maintaining correct FFT functionality and a minimum operating frequency of 10 MHz. Unlike the high-performance design, which prioritizes throughput and frequency scaling, the EE architecture focuses on minimizing dynamic switching activity, memory access, and unnecessary arithmetic operations within the FFT datapath. The accelerator energy decreases with a reduction in latency, dependent on cycles, and the total power consumed.

Two main inefficiencies can be identified with the baseline radix-2 DIT FFT: memory based twiddle access and recursive generation inside the butterfly loop. The memory based twiddle access increases SRAM activity and data movement, contributing to a large portion of power consumption. In contrast, recursive twiddle generation requires repeated complex multiplications, even when a small subset of twiddle factors is needed. This further contributes to a higher dynamic power consumption. Together, these inefficiencies also lead to unnecessary operations and steps that increase the overall cycle count of the FFT.

To address this, the EE takes advantage of the fact that the FFT size is fixed to 32 complex points and adopts a constant twiddle strategy. All required twiddle factors are embedded directly within the FFT datapath as fixed constants, eliminating the need for runtime twiddle generation and memory accesses from the critical execution path. This restricts all twiddle usage to local combinational logic, reducing the number of operations performed during execution.

In addition, trivial twiddle factors such as $+1$, -1, $+j$, and $-j$ were treated as special cases. These values do not require full complex multiplication and can instead be realized using sign inversion, real-imaginary swapping, and add/subtract operations. Exploiting these cases avoids unnecessary multiplier activity and reduces both switching and computational cost within the butterfly operations.

Furthermore, analysis of the baseline power reports revealed that the accelerator data memory is the dominant power-consuming block, accounting for over half of the total chip power. This is primarily due to the repeated SRAM reads and writes required to store and retrieve intermediate butterfly results after every butterfly operation.

To address this, all intermediate values are moved into an on-chip register file, similarly to the HP design. However, unlike the HP design where the motivation was port bandwidth, here one of the primary motivation is power reduction. This confines SRAM access to two bulk phases per chunk: a single LOAD phase that reads all input samples at the start, and a single STORE phase that writes all results at the end, with no SRAM activity during the compute phase.

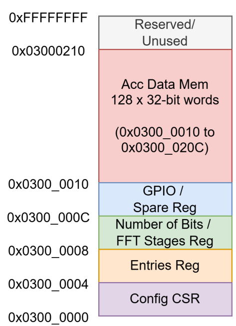


Fig. 4: Partial memory map of the EE accelerator, showing the control/configuration registers and internal accelerator SRAM. The register region spans 0x03000000 to 0x0300000C, while the 128-word accelerator data memory spans 0x03000010 to 0x0300020C.

As shown in Figure 4, the EE accelerator exposes a small MMIO register region for control and configuration, followed by an internal SRAM region starting at `0x03000010`. The control region contains the configuration/status register, the entry-count register, the number of bits/FFT stages register, and a spare GPIO-style register. The internal SRAM is used for bulk input loading and final output storage, which supports the design choice of restricting memory activity to the LOAD and STORE phases while keeping all intermediate FFT computation inside the local register file.

Together, these optimizations reduce the cycle count from 732 to 210 and lower dynamic power by eliminating the dominant source of switching activity. This forms the basis for algorithm level optimizations to reduce energy further, discussed next.

1) Design: We decided to optimize the algorithm to further reduce latency and consequently energy consumption. Firstly, the baseline loading phase loads all 64 values in 64 cycles (32 real parts and 32 imaginary parts), since the SRAM is accessed one word per cycle. However, the input signal contains only real components, meaning all imaginary components are always zero, resulting in 32 cycles wasted loading zeros. We therefore removed the loading of the 32 imaginary components, directly halving the load phase and reducing SRAM activity, which as established is the dominant source of power consumption.

The next step was to apply a Radix-4 decomposition to reduce the number of FFT stages. However, since $32 = 2^5$ is not a pure power of 4, a direct application results in a mixed decomposition: two Radix-4 stages followed by an unavoidable Radix-2 stage, yielding $8+8+16 = 32$ butterflies across 3 stages. This results in a significant cycle reduction when compared to the baseline (80 butterflies across 5 stages). By running the full simulation considering **only** the Radix-4-4-2 decomposition, a first chunk latency of $\approx 36\mu s$, total power of $\approx 0.62mW$ and thus a total energy of $\approx 22nJ$, were achieved. Despite a strong significant improvement in latency, total power increased but slightly beat the baseline energy. Therefore, an implementation of Clock gating to decrease power consumption combined with the improved latency can potentially reduce the energy consumption even more. Moreover, a pure Radix-4 decomposition would be more optimal to further reduce latency.

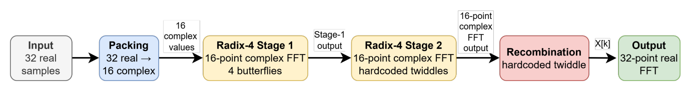


Fig. 5: Radix-4 RFFT Diagram

This leads us to the Radix 4 RFFT algorithm shown in Figure 5. Since the input signal is purely real, the 32 real valued samples can be packed into 16 complex values, reducing the problem to a 16-point FFT. As $16 = 4^2$, this decomposes into exactly two Radix-4 stages with a total of 8 butterflies stages and no remaining Radix-2 stage. The first stage involves only trivial twiddle factors $\{1, -j, -1, +j\}$), and therefore requires no multiplications, while the second stage uses pre-defined twiddle constants which are hard-coded into the data path. This design was also simulated, this time with the optimization, and found to have a first chunk latency of $\approx 10.8\mu s$, total power of $\approx 0.68mW$ and thus a total energy of $\approx 7.4nJ$. Despite the slight increase in power from the previous design, the latency is seen to have a large drop, resulting in a massive energy improvement from the previous design and baseline.

Since the RFFT two stage radix-4 demonstrated significant improvement form the baseline in terms of latency, the next step is to reduce the total power. For the final EE design presented, manual clock gating is performed for both the accelerator and its memory in order to preserve power. This is implemented by using the standard integrated clock gating

(ICG) `TLATNCAX2` cell from the `gpdk045` library. The cell works as follows: the enable signal is latched while the clock is low, ensuring it is stable before the clock rises. The gated clock output `ECK` is then the AND of the clock and the latched enable. Latching during the low phase prevents any glitches on the enable from propagating to `ECK`, producing a clean gated clock signal. This manual clock gating ensures that the accelerator is turned off when not in operation, reducing the overall power consumption. Manual clock gating was preferred over automatic clock gating synthesis, as the latter introduced hold violations that could not be resolved within the design constraints. This final EE design achieves a first-chunk latency of $10.83\,\mu$s, a total post-layout power of 0.239 mW, and thus an energy consumption of 2.59 nJ, representing a $9.5\times$ reduction compared to the 24.6 nJ baseline.

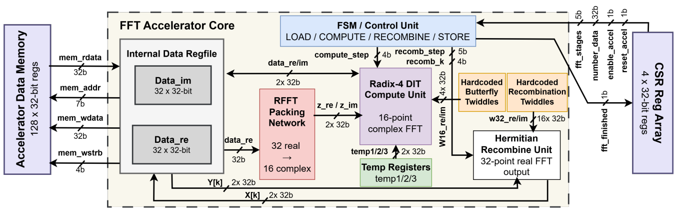


Fig. 6: RFFT-based accelerator core architecture with memory access - 16-point FFT using two Radix-4 stages. This includes real to complex packing, hardcoded twiddle factors, and a Hermitian recombination unit.

Our final design architecture that we use implements the discussed two stage Radix-4 RFFT in combination with the manual clock gating, presented in Figure 6.

B. Implementation

SDC Constraints and Rationale: The timing constraints were adapted from the baseline SDC to reflect the specific architectural characteristics of the Radix-4 RFFT accelerator, rather than applied unchanged. The clock period was kept at 83.33 ns (12 MHz) to match the baseline operating point. The global `set_max_transition` was increased from the baseline value of 0.20 ns to 0.28 ns. This decision was based on two structural properties of the design: the fully combinational RECOMBINE twiddle multiplication path, and the high-fanout nets produced by the parallel Stage 1 butterfly, both of which inherently exhibit higher net capacitance than the baseline datapath. Retaining the tighter 0.20 ns constraint introduces two timing risks. First, if violations are not resolved, slow signal transitions increase propagation delay and reduce setup slack. Second, if the tool aggressively fixes these violations by inserting buffers or upsizing drivers, path delays can become shorter, reducing hold margin, especially on short register-file paths. Relaxing the constraint to 0.28 ns avoids these issues while maintaining stable signal behavior.

Place-and-Route Tool Settings: In `6.cts.tcl`, there is one change compared to the baseline. The `holdTargetSlack` was increased from the baseline value of 0.1 ns to 0.2 ns. The `holdTargetSlack` sets a margin above zero that the tool must achieve for hold time paths. So instead of aiming for a +0.1 ns, the EE design implements a more aggressive hold fixing during clock tree synthesis, reducing the number of violations carried into later stages. Any remaining violations are resolved during the routing phase.

Therefore, an addtional `optDesign -postRoute -hold` step was added to `7.routing.tcl`. Prior to this step, `setOptMode -holdTargetSlack` was set to 0.05 ns, instructing the tool to target a small positive hold margin rather than zero. This ensures the optimizer inserts sufficient delay cells to close the residual violations with the margin, rather than stopping exactly at the hold boundary. These adjustments to the baseline was enough to fix the hold violations after the initial routing.

The final post-layout design is timing-clean, with no setup or hold violations in the post-route static timing analysis. The design is free from DRC and DRV violations, with the exception of `max_transition` on the high-fanout combinational nets described above, which are a known consequence of the architectural choice and are outside the scope of this course. All timing reports, DRC summaries, and place-and-route logs are included in the Appendix.

Floorplan & Congestion: The floorplan was kept the same as the baseline. The standard floorplan was enough for the given fixed area of 596.4 μm x 596.4 μm. In the end we reached an area of $268310.8\,\mu\text{m}^2$, which is smaller than the maximum specification.

C. Performance Metrics and Validation

After synthesis and the place and route, the final EE design power, performance and area metrics are presented in Table II. Seen on the table and section III-0b, the PNR is devoid of any timing violations. The final design has no DRC or DRV violations, except for `max_tran` (outside the scope of this course.), all highlighted in section III-0b. Furthermore, from the area report in section III-0b, the total area of the accelerator is shown. Although the total power is lower than the baseline, further reduction was expected through clock gating using Genus. However, this approach introduced significant hold violations that could not be resolved.

Post-Layout Simulation: Similar to the HP design, post-layout simulations were performed for both setup and hold timing. The final design passed all tests, and the FFT outputs matched the golden reference. Finally the `VCD` annotated power reports were generated and a 100% activity annotation was confirmed from the terminal, seen in Figure 11. Furthermore the cycles for every FSM state are counting and verified to be 130 cycles.

The final post-layout waveform simulations is displayed in section III-0b. A zoomed in version is also presented in Figure 12, where the `fft_finished` waveform declares that the accelerator is done with the fft after the `state_reg` reaches (11). Moreover, there is a small delay between the `clk` going to high and the `fft_finished` going to high.

TABLE II: Post-Layout Results

Metric	Value
Timing & Area	
Frequency	12 MHz
Core Area	268310.8 μm^2
Latency (cycles)	130
Latency (μs)	10.8329 μs
Sign-off	
Setup WNS	+35.091 ns
Hold WNS	+0.041 ns
DRC Violations	0
Critical DRV Violations	0
max_tran violations	795
Power (activity-annotated)	
Dynamic Power	0.053 mW
Leakage Power	0.029 mW
Total Power	0.239 mW
Energy ($P \times T$)	2.589 nJ
Annotation Coverage	100 %

A zoomed out waveform simulation is displayed in Figure 13, showing the complete `state_reg` transition flow.

To further confirm that the radix-4 rfft gives the same output as our software implemented version, the memory write is also checked. In Figure 14, the memory write waveform is shown. Comparing it to the golden reference which states that the last value should be equal to `-14 -5j`. These values are the same for the testbench and thus validate that the post-layout design is operating correctly.

III. Comparative PPA Trade-off Analysis

Table III summarises the post-layout PPA metrics for the baseline, HP, and EE accelerators.

TABLE III: Post-layout PPA comparison of the baseline, HP, and EE accelerators

Metric	Baseline	HP	EE
Frequency (MHz)	12	66.2	12
Core Area (μm^2)	355693.0	251293.9	268310.8
Latency (cycles)	732	121	130
Latency (μs)	61.00	1.8271	10.8329
Accel. Dynamic Power (mW)	0.093	1.652	0.001
Accel. Leakage Power (mW)	0.013	0.01667	0.020
Accel. Power (excl. RISC-V) (mW)	0.403	1.669	0.029
Total Power (mW)	0.626	2.795	0.239
Energy (nJ)	24.6	5.107	2.589
Setup WNS (ns)	+33.845	+0.226	+35.091
Hold WNS (ns)	+0.032	+0.015	+0.041
DRC Violations	0	0	0
Annotation Coverage (%)	100	100	100

The HP and EE designs represent a classic speed–power Pareto trade-off. Raising the clock $5.5\times$ buys the HP design a $33\times$ latency win at the cost of $4.5\times$ higher power, while the EE design's algorithmic restructuring (Radix-4 RFFT) and clock gating deliver the lowest energy at a moderate latency. Both designs independently beat the baseline energy of 24.6 nJ, confirming that cycle-count reduction is the dominant lever for energy efficiency regardless of the frequency strategy.

Highest-Impact Decisions: For the HP design, the wide paired memory port had the single largest cycle-count impact: even after dual butterfly units cut compute cycles in half, LOAD/STORE still accounted for 128 of 185 cycles ($\sim$69 %); doubling the memory bandwidth halved those phases and brought the total from 185 to 121 cycles. On the frequency axis, the 4-stage micro-pipeline was the enabling decision, breaking the $\sim$19.6 ns combinational butterfly path into four segments and allowing synthesis at 66.2 MHz with post-route closure.

For the EE design, the Radix-4 RFFT decomposition had the largest impact across all PPA dimensions simultaneously. Packing 32 real samples into 16 complex values reduced the problem to a two-stage Radix-4 FFT with only 8 butterflies (versus 80 in the baseline), cutting cycle count, eliminating non-trivial twiddle multiplications in Stage 1, and removing 32 wasted imaginary-zero load cycles. Manual clock gating via the `TLATNCAX2` ICG cell then amplified the power benefit by shutting down the accelerator and its memory when idle, reducing accelerator dynamic power to 0.001 mW.

Surprising Results and Lessons Learned: The most pervasive challenge shared by both designs was hold-timing closure during PnR. The HP design initially had 1,356 violating paths after Innovus's default hold-fixing pass; resolution required explicit hold-fixing cell lists, a $+20$ ps `holdTargetSlack`, and a dedicated post-route `optDesign` hold pass. The EE design encountered the same issue when Genus automatic clock gating introduced hold violations that could not be resolved, forcing a switch to manual ICG insertion via the `TLATNCAX2` cell. The key takeaway is that hold repair is an incremental, heuristic-driven process spanning multiple levels—from fine-grained buffer cell selection to high-level PnR strategy choices, and must be tailored to the design's specific characteristics rather than left to tool defaults.

Redesign Considerations: Given more time, combining the EE's Radix-4 RFFT algorithm with the HP's micro-pipeline and frequency scaling could yield a design that minimises both latency and energy simultaneously—an avenue not explored within the project timeline. On the EE side, finer-grained clock gating (e.g. per-stage register-file banks) could further suppress leakage during idle sub-phases of the computation. For the HP design, a wider $4\times$-parallel memory port or DMA-style burst transfers would further compress the LOAD/STORE phases that still constitute 53 % of the 121-cycle budget.

Appendix

A. High-Performance (HP) Design

A.1 Timing Reports:

Post-Route Setup Summary:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:52:48 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -pathReports -slackReports -numPaths 50 -outDir
↪  finalReports/report_timing
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.226  |  2.625  |  0.226  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|  12049  |  12044  |   313   |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |     10 (20)      |   -0.223   |     14 (28)      |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 69.138%
       (100.000% with Fillers)
Total number of glitch violations: 0
------------------------------------------------------------------
```

Post-Route Hold Summary:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:53:00 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -hold -pathReports -slackReports -numPaths 50
↪  -outDir finalReports/report_timing_hold
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.015  |  0.015  |  0.035  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|  12049  |  12044  |   313   |
+--------------------+---------+---------+---------+

Density: 69.138%
```

```
       (100.000% with Fillers)
------------------------------------------------------------------
```

A.2 DRC Reports:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:42:29 2026
#  Design:            et4351
#  Command:           verify_drc -limit 1000 -report verifyReports/verify_drc.rpt
###############################################################

No DRC violations were found
```

A.3 DRV Reports:

DRV summary: All DRV reports except DRV tran are clean.

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:53:45 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -drvReports -numPaths 500 -outDir
 ↪  finalReports/report_DRV
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.226  |  2.625  |  0.226  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|  12049  |  12044  |   313   |
+--------------------+---------+---------+---------+

+----------------+------------------------------+------------------+
|                |             Real             |       Total      |
|    DRVs        +------------------+-----------+------------------|
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  |
+----------------+------------------+-----------+------------------+
|   max_cap      |      0 (0)       |   0.000   |      0 (0)       |
|   max_tran     |     10 (20)      |  -0.223   |     14 (28)      |
|   max_fanout   |      0 (0)       |     0     |      0 (0)       |
|   max_length   |      0 (0)       |     0     |      0 (0)       |
+----------------+------------------+-----------+------------------+

Density: 69.138%
       (100.000% with Fillers)
Total number of glitch violations: 0
------------------------------------------------------------------
```

DRV tran violations:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:53:51 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -drvReports -numPaths 500 -outDir
 ↪  finalReports/report_DRV
###############################################################
# Net / InstPin                               MaxTranTime     TranTime        TranSlack
 ↪  CellPort              Remark
#
flash_io3
```

```
    soc/g2983/A0                          0.280r/0.280f    1.314r/1.580f    -1.034r/-1.300f
    ↪  AO21X1/A0                 M
    soc/spimemio/xfer/g3389/A0N           0.280r/0.280f    1.314r/1.580f    -1.034r/-1.300f
    ↪  OAI2BB1X1/A0N             M
flash_io2
    soc/g2986/A0                          0.280r/0.280f    1.310r/1.576f    -1.030r/-1.296f
    ↪  AO21X1/A0                 M
    soc/spimemio/xfer/g3388/A0N           0.280r/0.280f    1.310r/1.576f    -1.030r/-1.296f
    ↪  OAI2BB1X1/A0N             M
flash_io0
    soc/g2985/A0                          0.280r/0.280f    1.284r/1.544f    -1.004r/-1.264f
    ↪  AO21X1/A0                 M
    soc/spimemio/xfer/g3392/A0N           0.280r/0.280f    1.284r/1.544f    -1.004r/-1.264f
    ↪  OAI2BB1X1/A0N             M
flash_io1
    soc/g2980/A0                          0.280r/0.280f    1.213r/1.459f    -0.933r/-1.179f
    ↪  AO21X1/A0                 M
    soc/spimemio/xfer/drc_bufs3602/A      0.280r/0.280f    1.213r/1.459f    -0.933r/-1.179f
    ↪  BUFX2/A                   M
soc/cpu/FE_PHN17887_cpuregs_31__13
    soc/cpu/FE_PHC17887_cpuregs_31__13/Y     0.280r/0.280f    0.323r/0.503f    -0.043r/-0.223f
    ↪  BUFX3/Y                   R
    soc/cpu/FE_PDC21794_FE_PHN17887_cpuregs_31__13/A  0.280r/0.280f    0.319r/0.497f
    ↪  -0.038r/-0.217f  BUFX2/A                 R
soc/cpu/FE_PDN297_FE_PHN13174_cpuregs_15__14
    soc/cpu/FE_PDC297_FE_PHN13174_cpuregs_15__14/Y  0.280r/0.280f    0.307r/0.447f
    ↪  -0.027r/-0.167f  CLKINVX4/Y              R
    soc/cpu/g92606/B                      0.280r/0.280f    0.306r/0.445f    -0.026r/-0.165f
    ↪  MXI2XL/B                  R
soc/cpu/FE_PDN288_FE_PHN15539_cpuregs_9__2
    soc/cpu/FE_PDC288_FE_PHN15539_cpuregs_9__2/Y  0.280r/0.280f    0.226r/0.383f
    ↪  0.054r/-0.103f   INVX6/Y                 R
    soc/cpu/FE_PDC21782_FE_PHN15539_cpuregs_9__2/A  0.280r/0.280f    0.224r/0.379f
    ↪  0.056r/-0.099f   BUFX2/A                 R
accel/mem/n_182
    accel/mem/g99626/Y                    0.280r/0.280f    0.276r/0.368f    0.004r/-0.088f
    ↪  INVX6/Y                   R
    accel/mem/FE_PDC21746_n_182/A         0.280r/0.280f    0.273r/0.364f    0.007r/-0.084f
    ↪  BUFX2/A                   R
accel/mem/FE_PDN21763_n_199
    accel/mem/FE_PDC21763_n_199/Y         0.280r/0.280f    0.230r/0.343f    0.050r/-0.063f
    ↪  BUFX4/Y                   R
    accel/mem/FE_PDC21796_n_199/A         0.280r/0.280f    0.228r/0.339f    0.052r/-0.059f
    ↪  BUFX2/A                   R
soc/cpu/FE_PDN21793_n
    soc/cpu/FE_PDC21793_n/Y               0.280r/0.280f    0.225r/0.333f    0.055r/-0.053f
    ↪  BUFX3/Y                   R
    soc/cpu/FE_PDC406_n/A                 0.280r/0.280f    0.224r/0.331f    0.056r/-0.051f
    ↪  INVX2/A                   R
soc/cpu/cpuregs_2__7_
    soc/cpu/FE_PHC8838_cpuregs_2__7/A     0.280r/0.280f    0.241r/0.313f    0.039r/-0.033f
    ↪  CLKBUFX2/A                R
    soc/cpu/cpuregs_reg_2__7_/Q           0.280r/0.280f    0.241r/0.313f    0.039r/-0.033f
    ↪  DFFHQX2/Q                 R
accel/FE_OFN1944_cpu_mem_rdata_9
    accel/FE_OFC1944_cpu_mem_rdata_9/Y    0.280r/0.280f    0.249r/0.304f    0.031r/-0.024f
    ↪  CLKINVX6/Y                R
    accel/FE_OFC2241_cpu_mem_rdata_9/A    0.280r/0.280f    0.244r/0.297f    0.036r/-0.017f
    ↪  INVX2/A                   R
soc/cpu/cpuregs_24__0_
    soc/cpu/FE_PHC16766_cpuregs_24__0/A   0.280r/0.280f    0.230r/0.287f    0.050r/-0.007f
    ↪  CLKBUFX2/A                R
    soc/cpu/cpuregs_reg_24__0_/Q          0.280r/0.280f    0.230r/0.287f    0.050r/-0.007f
    ↪  DFFHQX2/Q                 R
    soc/cpu/g95827/A0                     0.280r/0.280f    0.230r/0.287f    0.050r/-0.007f
    ↪  AOI22X1/A0                R
soc/cpu/reg_op1[15]
```

```
    soc/cpu/reg_op1_reg_15_/Q                0.280r/0.280f    0.217r/0.281f    0.063r/-0.001f
     ↪  DFFX4/Q                          R

*info: there are 28 max_tran violations in the design.
*info: 20 violations are real (remark R).
*info: 8 violations may not be fixable:
*info:     8 violations on multiple fanin net (remark M).
```

A.4 Connectivity Reports:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:42:23 2026
#  Design:            et4351
#  Command:           verifyConnectivity -type all -error 1000 -warning 50 -report
 ↪  verifyReports/verifyConnectivity.rpt
###############################################################
Verify Connectivity Report is created on Tue Apr  7 23:42:23 2026



Multi-CPU acceleration using 4 CPU


Use pthread


Begin Summary
    Found no problems or warnings.
End Summary
```

A.5 Geometry Reports:

Area Report:

```
    Hinst Name              Module Name
    ↪  Inst Count           Total Area
-------------------------------------------------------------------------------------------------------------------↲
↪  -------------------------------------------------------------------------------------------------------------
et4351
↪  65307           251293.939
 accel                      accelerator
 ↪  41976           140132.448
  accel/fft                 accelerator_fft_LOG_MAX_N32_MEM_WIDTH24_ADDR_WIDTH6_NUM_TW16
  ↪  27227           91371.114
  accel/mem                 accelerator_mem_MEM_DEPTH64_DATA_WIDTH24
  ↪  9137            31334.382
 flash_bufs_0_              bidir_buf_2679
 ↪  4               25.308
 flash_bufs_1_              bidir_buf_2680
 ↪  3               19.494
 flash_bufs_2_              bidir_buf_2681
 ↪  4               29.412
 flash_bufs_3_              bidir_buf
 ↪  4               29.412
 soc                        picosoc
 ↪  23265           110752.117
  soc/cpu                   picorv32_ENABLE_COUNTERS1_BARREL_SHIFTER1_COMPRESSED_ISA1_ENABLE_MUL1_ENABLE_FAST_MUL0_ENABLE↲
  ↪  _DIV1_ENABLE_IRQ1_ENABLE_IRQ_QREGS0_PROGADDR_RESET1048576_PROGADDR_IRQ0_STACKADDR1024      20612
  ↪  66018.996
   soc/cpu/genblk1.pcpi_mul picorv32_pcpi_mul
   ↪  1444           6662.844
   soc/cpu/genblk2.pcpi_div picorv32_pcpi_div
   ↪  1502           6205.932
  soc/memory                picosoc_mem_WORDS256
  ↪  24           34581.193
  soc/simpleuart            simpleuart
  ↪  1028           4172.400
  soc/spimemio              spimemio
  ↪  1147           4683.006
   soc/spimemio/xfer        spimemio_xfer
   ↪  355           1449.054
```

Geometry Verification Summary:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:42:29 2026
#  Design:            et4351
```

```
#  Command:           verify_drc -limit 1000 -report verifyReports/verify_drc.rpt
###############################################################

No DRC violations were found
```

Placement Check:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Mon Apr  6 12:22:34 2026
#  Design:            et4351
#  Command:           checkPlace verifyReports/checkPlace.rpt
###############################################################

## No violations found ##

## Summary:
#########################################################
## Number of Placed Instances = 46682
##   of which 4 are fixed.
## Number of Unplaced Instances = 0
## Placement Density:42.58%(133485/313515)
## Placement Density (including fixed std cells):42.58%(133485/313515)
```

Floorplan Check:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Mon Apr  6 12:11:34 2026
#  Design:            et4351
#  Command:           checkFPlan -outFile verifyReports/checkFPlan.rpt
###############################################################
WARNING (IMPFP-7236): DIE's corner: (664.8000000000 , 664.8000000000) is NOT on PlacementGrid,  Please use command
↪  getFPlanMode to check current Grid settings, use command setFPlanMode to change which grid to snap to. And use command
↪  get_snap_grid_info to get grids' offset and pitch. Command floorplan can be used to fix this issue.
WARNING (IMPFP-7238): CORE's  corner:        (630.6000000000 , 630.6000000000)  is        NOT on PlacementGrid. You can
↪  use the getFPlanMode command to check the current grid settings and use the setFPlanMode command to  change the grid
↪  to snap to. You can also use the get_snap_grid_info command to get information on the offset and pitch of the current
↪  snap grid. To resolve this issue, use the floorPlan command.
INFO (IMPFP-7241): Tech site CoreSiteDouble has no symmetry Y.         Current symmetry setting : isSymmetryR90 false;
↪  isSymmetryX : false; isSymmetryY : false;

INFO (IMPFP-7241): Tech site CoreSite has no symmetry Y.         Current symmetry setting : isSymmetryR90 false;
↪  isSymmetryX : false; isSymmetryY : false;

WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_0 and snapping rules are not checked since it's
↪  width is not multiple of default tech site's width.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_0 and snapping rules are not checked since it's
↪  height is not multiple of default tech site's height.
INFO (IMPFP-10122): Instance soc/memory/sram_0 not snapped to row-site.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_1 and snapping rules are not checked since it's
↪  width is not multiple of default tech site's width.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_1 and snapping rules are not checked since it's
↪  height is not multiple of default tech site's height.
INFO (IMPFP-10122): Instance soc/memory/sram_1 not snapped to row-site.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_2 and snapping rules are not checked since it's
↪  width is not multiple of default tech site's width.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_2 and snapping rules are not checked since it's
↪  height is not multiple of default tech site's height.
INFO (IMPFP-10122): Instance soc/memory/sram_2 not snapped to row-site.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_3 and snapping rules are not checked since it's
↪  width is not multiple of default tech site's width.
WARNING (IMPFP-10013): Halo should be created around block soc/memory/sram_3 and snapping rules are not checked since it's
↪  height is not multiple of default tech site's height.
INFO (IMPFP-10122): Instance soc/memory/sram_3 not snapped to row-site.
**INFO: No. of regular pre-routes not on tracks : 0
```

A.6 Antenna Reports:

a) Antenna Verification Report:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Tue Apr  7 23:45:17 2026
#  Design:            et4351
#  Command:           verifyProcessAntenna -error 1000 -reportfile verifyReports/verifyProcessAntenna.rpt
###############################################################

No Violations Found
```

A.7 Activity Annotation Reports:

```
  Filename (activity)                      :
../sim_phys/vcd/et4351.phys.hold.vcd
  Names in file that matched to design    : 67937/67946
  Annotation coverage for this file
   (Unique nets matched/Total nets)        : 67819/67819 = 100%

  9 nets were found in the VCD file(s) but were not in
  the design.  These nets are listed in the file
voltus_power_missing_netnames.rpt.

  Total annotation coverage for all files of type VCD: 67819/67819 = 100%
  Percent of VCD annotated nets with zero toggles: 64056/67819 = 94.4514%
```

Fig. 7: Coverage report after physical simulation

b) Annotation Coverage:

Initial Activity Annotation Check:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Mon Apr  6 12:11:34 2026
#  Design:            et4351
#  Command:           report_annotated_check > initialReports/report_annotated_check.rpt
###############################################################
     +----------------------------------------------------------------------+
     |     Check arc type       | Total | Annotated |    Not     | Percentage |
     |                          |       |           | Annotated |    Arcs    |
     |                          |       |           |            |  Annotated |
     |--------------------------+-------+-----------+------------+------------|
     |       Setup arcs         | 11865 |     0     |   11865    |  0.000000  |
     |      Recovery arcs       |  192  |     0     |    192     |  0.000000  |
     |  Min pulse width arcs    | 14326 |     0     |   14326    |  0.000000  |
     | Clock gating setup arcs  |   4   |     0     |     4      |  0.000000  |
     +--------------------------+-------+-----------+------------+------------+
     |         All arcs         | 26387 |     0     |   26387    |  0.000000  |
     +----------------------------------------------------------------------+
```

Pre-Place VCD Import Report:

```
*----------------------------------------------------------------------------------------
*        Innovus 21.11-s130_1 (64bit) 08/19/2021 16:59 (Linux 3.10.0-693.el7.x86_64)
*
*
*         Date & Time:        2026-Apr-06 12:11:56 (2026-Apr-06 10:11:56 GMT)
*
*----------------------------------------------------------------------------------------
*
*        Design: et4351
*
*        Liberty Libraries used:
*                analysis_view_power:
↪   /data/Cadence/gpdk045_v60/gsclib045_svt_v4.7/gsclib045/timing/fast_vdd1v2_basicCells.lib
*                analysis_view_power: /data/Cadence/gpdk045_v60/Synopsys_sram/saed32sram_tt1p05v125c.lib
*
*        Power Domain used:
*              Rail:        VDD      Voltage:       1.32
*
*       Power View : analysis_view_power
*
*       User-Defined Activity : N.A.
*
*        Switching Activity File used:
*                ../sim_struct/vcd/et4351.struct.vcd
*                    Vcd Window used(Start Time, Stop Time):
*                     Vcd Scale Factor: 1
* *                    Design annotation coverage: 0/50916 = 0%
*
*       Hierarchical Global Activity: N.A.
*
*       Global Activity: N.A.
*
*       Sequential Element Activity: N.A.
*
*       Primary Input Activity: N.A.
*
*       Default icg ratio: N.A.
*
*       Global Comb ClockGate Ratio: N.A.
```

```
*
*        Power Units = 1mW
*
*        Time Units = 1e-09 secs
*
*       report_power -outfile powerReports/prePlace_VCDImport.rpt
*
-----------------------------------------------------------------------------------------


Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        0.34310520           93.7134%
Total Switching Power:       0.00281524            0.7689%
Total Leakage Power:         0.02020141            5.5177%
Total Power:                 0.36612186
-----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Sequential                        0.3392   0.0003732     0.00619      0.3458       94.45
Macro                           0.000322   1.978e-06           0    0.000324     0.08849
IO                                     0           0           0           0           0
Combinational                   0.003551     0.00244     0.01401        0.02       5.463
Clock (Combinational)                  0           0           0           0           0
Clock (Sequential)                     0           0           0           0           0
-----------------------------------------------------------------------------------------
Total                             0.3431    0.002815      0.0202      0.3661         100
-----------------------------------------------------------------------------------------


Rail                  Voltage   Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
VDD                      1.32     0.3431    0.002815      0.0202      0.3661         100


-----------------------------------------------------------------------------------------
*        Power Distribution Summary:
*                Highest Average Power:        soc/spimemio/g9085 (CLKMX2X8):        0.0006393
*                Highest Leakage Power:         accel/fft/g182806 (BUFX20):        2.293e-06
*          Total Cap:      7.43277e-11 F
*          Total instances in design: 48224
*          Total instances in design with no power:     0
*          Total instances in design with no activity:     0
*          Total Fillers and Decap:     0
-----------------------------------------------------------------------------------------
```

Post-Route VCD-Based Power Report:

```
*----------------------------------------------------------------------------------------
*        Innovus 21.11-s130_1 (64bit) 08/19/2021 16:59 (Linux 3.10.0-693.el7.x86_64)
*
*
*         Date & Time:        2026-Apr-08 00:32:57 (2026-Apr-07 22:32:57 GMT)
*
*----------------------------------------------------------------------------------------
*
*        Design: et4351
*
*        Liberty Libraries used:
*                analysis_view_power:
↪  /home/nfs/lcastello/project/pnr/checkpoints/et4351_done.enc.dat/libs/mmmc/fast_vdd1v2_basicCells.lib
*                analysis_view_power:
↪  /home/nfs/lcastello/project/pnr/checkpoints/et4351_done.enc.dat/libs/mmmc/saed32sram_tt1p05v125c.lib
*
*        Power Domain used:
*              Rail:        VDD      Voltage:       1.32
*
*       Power View : analysis_view_power
*
*       User-Defined Activity : N.A.
*
*        Switching Activity File used:
*                ../sim_phys/vcd/et4351.phys.hold.vcd
*                    Vcd Window used(Start Time, Stop Time):
*                     Vcd Scale Factor: 1
* *                    Design annotation coverage: 0/67999 = 0%
*
*       Hierarchical Global Activity: N.A.
*
*       Global Activity: N.A.
```

```
*
*       Sequential Element Activity: 0.200000
*
*       Primary Input Activity: N.A.
*
*       Default icg ratio: N.A.
*
*       Global Comb ClockGate Ratio: N.A.
*
*        Power Units = 1mW
*
*        Time Units = 1e-09 secs
*
*       report_power -outfile finalReports/report_power_postRouteVCD.rpt -hierarchy all
*
-----------------------------------------------------------------------------------------


Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        2.12477631           76.0171%
Total Switching Power:       0.64451861           23.0586%
Total Leakage Power:         0.02583624            0.9243%
Total Power:                 2.79513115
-----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Sequential                         1.825    0.002894     0.00592       1.833       65.59
Macro                          2.716e-05   9.136e-05           0   0.0001185     0.00424
IO                                     0           0   2.028e-07   2.028e-07   7.256e-06
Combinational                    0.02914     0.03199      0.0197     0.08082       2.892
Clock (Combinational)              0.271      0.6095   0.0002204      0.8808       31.51
Clock (Sequential)                     0           0           0           0           0
-----------------------------------------------------------------------------------------
Total                              2.125      0.6445     0.02584       2.795         100
-----------------------------------------------------------------------------------------


Clock                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
clk                                0.271      0.6095   0.0002204      0.8808       31.51
-----------------------------------------------------------------------------------------
Total (excluding duplicates)       0.271      0.6095   0.0002204      0.8808       31.51
-----------------------------------------------------------------------------------------
Clock: clk
Clock Period: 0.015100 usec
Clock Toggle Rate:   124.7947 Mhz
Clock Static Probability:  0.4992


Rail                 Voltage   Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
VDD                     1.32       2.125      0.6445     0.02584       2.795         100


Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        2.12477631           76.0171%
Total Switching Power:       0.64451861           23.0586%
Total Leakage Power:         0.02583624            0.9243%
Total Power:                 2.79513115
-----------------------------------------------------------------------------------------


Hierarchy                       Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
soc                                0.738      0.1911    0.009091      0.9381       33.56
accel                               1.33       0.322     0.01667       1.669       59.71
flash_bufs_0_                          0           0     2.78e-06    2.78e-06   9.947e-05
flash_bufs_1_                          0   0.0006451   1.872e-06    0.000647     0.02315
flash_bufs_2_                          0           0   2.776e-06   2.776e-06   9.931e-05
flash_bufs_3_                          0           0   2.776e-06   2.776e-06   9.931e-05
soc/cpu                           0.6011      0.1378    0.007933      0.7468       26.72
soc/memory                     5.346e-05   0.0002012   5.795e-06   0.0002605    0.009319
soc/spimemio                     0.06912      0.0163   0.0005162     0.08593       3.074
accel/fft                         0.7021      0.1694       0.011      0.8825       31.57
accel/mem                         0.4394     0.08997     0.00363       0.533       19.07
soc/simpleuart                   0.05276    0.004707   0.0004654     0.05793       2.073
soc/cpu/genblk2.pcpi_div         0.05196   1.745e-05   0.0006857     0.05266       1.884
```

```
soc/spimemio/xfer                0.02459    0.009123   0.0001565     0.03387       1.212
soc/cpu/genblk1.pcpi_mul          0.0682    0.006834   0.0007199     0.07575        2.71


-----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*               Highest Average Power:       CTS_ccl_a_buf_00007 (CLKBUFX20):          0.00989
*               Highest Leakage Power: accel/fft/FE_OFC608_n_8579 (CLKBUFX20):        2.292e-06
*         Total Cap:      2.53784e-10 F
*         Total instances in design: 65307
*         Total instances in design with no power:     0
*         Total instances in design with no activity:     0
*         Total Fillers and Decap:     0
-----------------------------------------------------------------------------------------
```

A.8 Power Analysis Reports:

Final Hierarchical Power Report:

```
*----------------------------------------------------------------------------------------
*       Innovus 21.11-s130_1 (64bit) 08/19/2021 16:59 (Linux 3.10.0-693.el7.x86_64)
*
*
*        Date & Time:        2026-Apr-07 23:52:16 (2026-Apr-07 21:52:16 GMT)
*
*----------------------------------------------------------------------------------------
*
*       Design: et4351
*
*       Liberty Libraries used:
*               analysis_view_power:
↪  /home/nfs/lcastello/project/pnr/checkpoints/et4351_cts.enc.dat/libs/mmmc/fast_vdd1v2_basicCells.lib
*               analysis_view_power:
↪  /home/nfs/lcastello/project/pnr/checkpoints/et4351_cts.enc.dat/libs/mmmc/saed32sram_tt1p05v125c.lib
*
*       Power Domain used:
*             Rail:        VDD      Voltage:       1.32
*
*      Power View : analysis_view_power
*
*      User-Defined Activity : N.A.
*
*      Activity File: N.A.
*
*      Hierarchical Global Activity: N.A.
*
*      Global Activity: N.A.
*
*      Sequential Element Activity: 0.200000
*
*      Primary Input Activity: N.A.
*
*      Default icg ratio: N.A.
*
*      Global Comb ClockGate Ratio: N.A.
*
*       Power Units = 1mW
*
*       Time Units = 1e-09 secs
*
*      report_power -outfile finalReports/report_power.rpt -hierarchy all
*
-----------------------------------------------------------------------------------------


Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        0.40996102           73.2244%
Total Switching Power:       0.12415566           22.1758%
Total Leakage Power:         0.02575247            4.5997%
Total Power:                 0.55986915
-----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
Sequential                        0.3507    0.000537    0.005918      0.3572        63.8
Macro                          2.441e-06   1.868e-05           0   2.112e-05    0.003772
IO                                     0           0   2.028e-07   2.028e-07   3.622e-05
Combinational                   0.007206     0.00748     0.01961      0.0343       6.127
Clock (Combinational)            0.05202      0.1161   0.0002204      0.1684       30.07
Clock (Sequential)                     0           0           0           0           0
-----------------------------------------------------------------------------------------
Total                               0.41      0.1242     0.02575      0.5599         100
-----------------------------------------------------------------------------------------
```

```
Clock                            Internal   Switching     Leakage       Total  Percentage
                                 Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
clk                               0.05202      0.1161   0.0002204      0.1684       30.07
-----------------------------------------------------------------------------------------
Total (excluding duplicates)      0.05202      0.1161   0.0002204      0.1684       30.07
-----------------------------------------------------------------------------------------
Clock: clk
Clock Period: 0.015100 usec
Clock Toggle Rate:    23.9682 Mhz
Clock Static Probability:  0.4997


Rail                  Voltage    Internal   Switching     Leakage       Total  Percentage
                                 Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
VDD                      1.32        0.41      0.1242     0.02575      0.5599         100


Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        0.40996102            73.2244%
Total Switching Power:       0.12415566            22.1758%
Total Leakage Power:         0.02575247             4.5997%
Total Power:                 0.55986915
-----------------------------------------------------------------------------------------


Hierarchy                        Internal   Switching     Leakage       Total  Percentage
                                 Power      Power         Power         Power  (%)
-----------------------------------------------------------------------------------------
soc                                0.1439     0.03788    0.009098      0.1909       34.09
accel                              0.2553     0.06142     0.01658      0.3333       59.52
flash_bufs_0_                           0           0    2.78e-06    2.78e-06   0.0004966
flash_bufs_1_                           0   0.0001184   1.872e-06   0.0001202     0.02148
flash_bufs_2_                           0           0   2.776e-06   2.776e-06   0.0004958
flash_bufs_3_                           0           0   2.776e-06   2.776e-06   0.0004958
soc/cpu                            0.1161     0.02686     0.00794      0.1509       26.95
soc/memory                       8.11e-06   3.829e-05   5.772e-06   5.217e-05    0.009319
soc/spimemio                      0.01466    0.003906   0.0005167     0.01908       3.409
accel/fft                          0.1348     0.03236      0.0109      0.1781       31.81
accel/mem                         0.08412     0.01701    0.003632      0.1048       18.71
soc/simpleuart                    0.01028   0.0009952   0.0004648     0.01174       2.097
soc/cpu/genblk2.pcpi_div          0.00998   4.754e-06   0.0006861     0.01067       1.906
soc/spimemio/xfer                0.006014    0.002429   0.0001565    0.008599       1.536
soc/cpu/genblk1.pcpi_mul          0.01305     0.00128   0.0007186     0.01504       2.687


-----------------------------------------------------------------------------------------
*        Power Distribution Summary:
*                Highest Average Power:       CTS_ccl_a_buf_00007 (CLKBUFX20):         0.001899
*                Highest Leakage Power: accel/fft/FE_OFC608_n_8579 (CLKBUFX20):        2.292e-06
*          Total Cap:      2.51931e-10 F
*          Total instances in design: 65307
*          Total instances in design with no power:     0
*          Total instances in design with no activity:     0
*          Total Fillers and Decap:     0
-----------------------------------------------------------------------------------------
```

Post-Route VCD-Based Power Report: see Post-Route VCD-Based Power Report in App.III-0a.

A.9 Sign-off Simulation Waveforms:

Post-Layout Setup Waveform: The sign-off setup waveform can be found in Fig.8, Fig.9, and Fig.10.

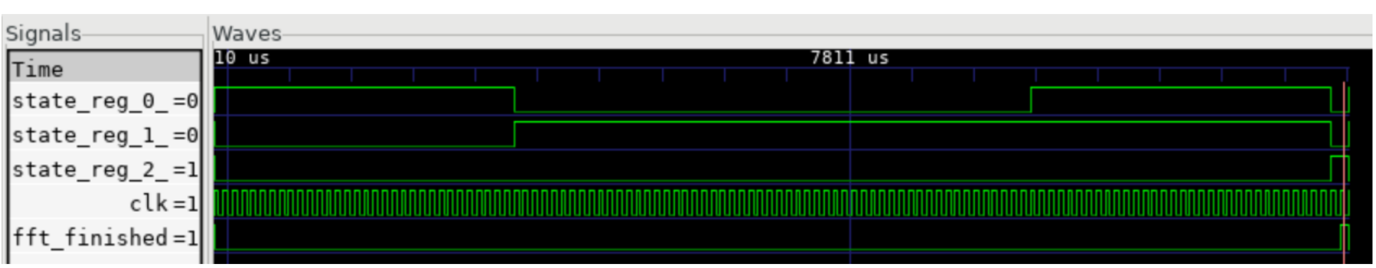


Fig. 8: Complete runtime waveform, starting from the data loading stage(001) and ending at the finish stage(100).

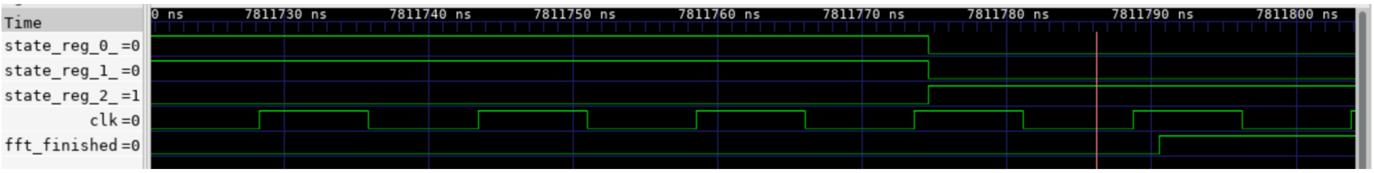


Fig. 9: Zoomed-in waveform illustrating the finished state, where `fft_finished` transitions from 0 to 1, and the stage ends at the finish state(100).

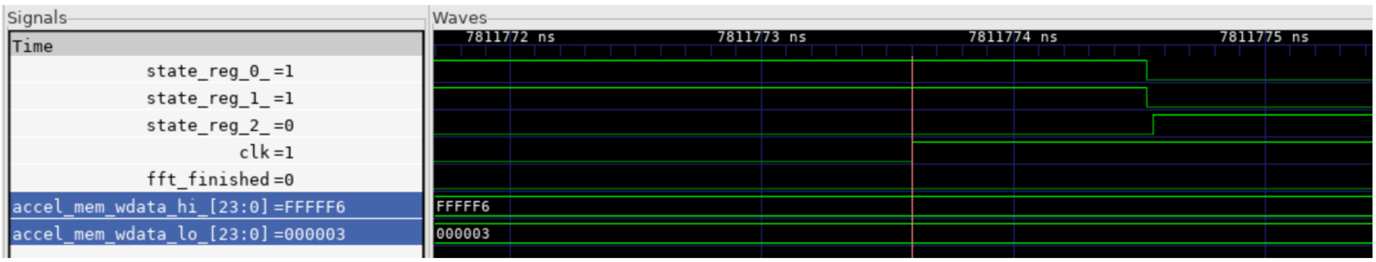


Fig. 10: Waveform showing the final output, which matches the expected value $3 - 10j$. Here, `accel_mem_wdata_hi` represents the imaginary part and `accel_mem_wdata_lo` represents the real part.

Post-Layout Hold Waveform: The sign-off hold waveform is functionally the same, and it look almost identical to the waveforms shown in Fig.8, Fig.9, and Fig.10, so it is not included in the appendix.

A.10 Post-Layout Test Specifications

Table IV lists the elements validated by post-layout simulation at both timing corners. All tests passed for the HP design.

TABLE IV: HP Post-Layout Test Specifications and Validation Results

ID	**Test Description**	**Result**
T1	FFT output data matches golden reference model	Pass
T2	Latency verification (Target = 121 cycles)	Pass
T3	FSM: `INIT` → `LOAD` → `COMPUTE` → `STORE` → `FINISH`	Pass
T4	Twiddle CSR loaded before enable	Pass
T5	Multi-chunk audio processing (24 chunks)	Pass
T6	Setup corner (max): No X-propagation	Pass
T7	Hold corner (min): No data corruption	Pass
T8	100% VCD annotation coverage reached	Pass

B. Energy-Efficient (EE) Design

B.1 Timing Reports:

Post-Route Setup Summary:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Fri Apr 10 00:05:13 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -pathReports -slackReports -numPaths 50 -outDir
↪  finalReports/report_timing
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| default |
+--------------------+---------+---------+---------+---------+
|           WNS (ns):| 35.299  | 38.994  | 78.749  | 35.299  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |
|          All Paths:| 13585   | 13578   |    2    |   226   |
+--------------------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |    316 (862)     |   -0.668   |    320 (870)     |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 74.566%
       (100.000% with Fillers)
Total number of glitch violations: 2
------------------------------------------------------------------
```

Post-Route Hold Summary:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Fri Apr 10 00:05:26 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -hold -pathReports -slackReports -numPaths 50
↪  -outDir finalReports/report_timing_hold
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg |reg2cgate| default |
+--------------------+---------+---------+---------+---------+
|           WNS (ns):|  0.056  |  0.056  |  0.243  |  0.100  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |
|          All Paths:| 13243   | 13236   |    2    |   226   |
+--------------------+---------+---------+---------+---------+

Density: 74.566%
       (100.000% with Fillers)
------------------------------------------------------------------
```

B.2 DRC Reports:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Wed Apr  8 21:36:22 2026
#  Design:            et4351
#  Command:           verify_drc -limit 1000 -report verifyReports/verify_drc.rpt
###############################################################

No DRC violations were found
```

B.3 DRV Reports:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Fri Apr 10 00:06:18 2026
#  Design:            et4351
#  Command:           timeDesign -postRoute -drvReports -numPaths 500 -outDir
 ↪  finalReports/report_DRV
###############################################################

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

+--------------------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg |reg2cgate| default |
+--------------------+---------+---------+---------+---------+
|           WNS (ns):| 35.299  | 38.994  | 78.749  | 35.299  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |
|          All Paths:|  13585  |  13578  |    2    |   226   |
+--------------------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |    316 (862)     |   -0.668   |    320 (870)     |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 74.566%
       (100.000% with Fillers)
Total number of glitch violations: 2
------------------------------------------------------------------
```

B.4 Connectivity Reports:

```
###############################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Wed Apr  8 21:36:22 2026
#  Design:            et4351
#  Command:           verify_drc -limit 1000 -report verifyReports/verify_drc.rpt
###############################################################

No DRC violations were found
```

B.5 Geometry Reports:

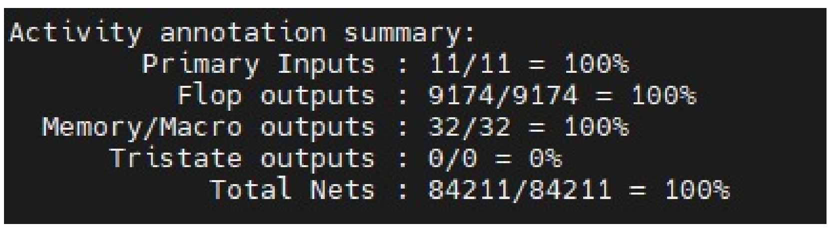
```
    Hinst Name                Module Name
    ↪  Inst Count              Total Area
---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------⌋
↪  ------------------------------------------------------------------
et4351
↪  73849          268310.833
 accel                        accelerator
 ↪  50900          162304.992
  accel/fft                   accelerator_fft_LOG_MAX_N32_MEM_WIDTH32_ADDR_WIDTH7
  ↪  38308          110843.226
```

```
 accel/mem                 accelerator_mem_MEM_DEPTH128
↪  11030            46571.508
 flash_bufs_0_             bidir_buf_4952
↪  2              17.784
 flash_bufs_1_             bidir_buf_4953
↪  2              17.784
 flash_bufs_2_             bidir_buf_4954
↪  2              17.784
 flash_bufs_3_             bidir_buf
↪  2              17.784
 soc                       picosoc
↪  22908          105793.459
 soc/cpu                   picorv32_ENABLE_COUNTERS1_BARREL_SHIFTER1_COMPRESSED_ISA1_ENABLE_MUL1_ENABLE_FAST_MUL0_ENABLE_DIV1_ENABLE_IRQ1_ENABLE_IRQ_QREGS0_PROGADDR_RESE
↪  T1048576_PROGADDR_IRQ0_STACKADDR1024      20024           61503.228
  soc/cpu/genblk1.pcpi_mul picorv32_pcpi_mul
 ↪  1259             5870.772
  soc/cpu/genblk2.pcpi_div picorv32_pcpi_div
 ↪  1866             5821.182
 soc/memory                picosoc_mem_WORDS256
↪  33            34622.575
 soc/simpleuart            simpleuart
↪  1205             4130.676
 soc/spimemio              spimemio
↪  1339             4775.346
  soc/spimemio/xfer        spimemio_xfer
 ↪  410              1403.226
```

B.6 Antenna Reports:

```
###################################################################
#  Generated by:      Cadence Innovus 21.11-s130_1
#  OS:                Linux x86_64(Host ID et4351.ewi.tudelft.nl)
#  Generated on:      Wed Apr  8 21:39:22 2026
#  Design:            et4351
#  Command:           verifyProcessAntenna -error 1000 -reportfile
↪  verifyReports/verifyProcessAntenna.rpt
###################################################################

No Violations Found
```



```
Activity annotation summary:
          Primary Inputs : 11/11 = 100%
            Flop outputs : 9174/9174 = 100%
  Memory/Macro outputs : 32/32 = 100%
        Tristate outputs : 0/0 = 0%
              Total Nets : 84211/84211 = 100%
```

Fig. 11: Screenshot of the activity annotation printed in the terminal

B.7 Activity Annotation Reports:

B.8 Power Analysis Reports:

```
*-----------------------------------------------------------------------------------------
*        Innovus 21.11-s130_1 (64bit) 08/19/2021 16:59 (Linux 3.10.0-693.el7.x86_64)
*
*
*         Date & Time:        2026-Apr-10 00:13:09 (2026-Apr-09 22:13:09 GMT)
*
*-----------------------------------------------------------------------------------------
*
*        Design: et4351
*
*        Liberty Libraries used:
*                analysis_view_power: /home/nfs/asakr/project/design_final/pnr/checkpoints/et4351_done.enc
↪  .dat/libs/mmmc/fast_vdd1v2_basicCells.lib
*                analysis_view_power: /home/nfs/asakr/project/design_final/pnr/checkpoints/et4351_done.enc
↪  .dat/libs/mmmc/saed32sram_tt1p05v125c.lib
*
*        Power Domain used:
*              Rail:        VDD      Voltage:       1.32
*
*       Power View : analysis_view_power
*
*       User-Defined Activity : N.A.
*
*        Switching Activity File used:
*                ../sim_phys/vcd/et4351.phys.hold.vcd
*                    Vcd Window used(Start Time, Stop Time):
*                     Vcd Scale Factor: 1
* *                    Design annotation coverage: 0/79559 = 0%
```

```
*
*       Hierarchical Global Activity: N.A.
*
*       Global Activity: N.A.
*
*       Sequential Element Activity: N.A.
*
*       Primary Input Activity: N.A.
*
*       Default icg ratio: N.A.
*
*       Global Comb ClockGate Ratio: N.A.
*
*        Power Units = 1mW
*
*        Time Units = 1e-09 secs
*
*       report_power -outfile finalReports/report_power_postRouteVCD.rpt -hierarchy all
*
-----------------------------------------------------------------------------------------
```

Total Power

Total Internal Power:	0.15620897	65.4360%
Total Switching Power:	0.05339382	22.3667%
Total Leakage Power:	0.02911743	12.1973%
Total Power:	0.23872022	

Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
Sequential	0.1293	0.0005052	0.007157	0.1369	57.36
Macro	0	1.964e-05	3.485e-12	1.964e-05	0.008226
IO	0	0	3.18e-07	3.18e-07	0.0001332
Combinational	0.006076	0.006365	0.0217	0.03414	14.3
Clock (Combinational)	0.02047	0.0465	0.0002579	0.06723	28.16
Clock (Sequential)	0.0003894	0	3.434e-06	0.0003928	0.1645
Total	0.1562	0.05339	0.02912	0.2387	100

Clock	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
clk	0.02086	0.0465	0.0002613	0.06762	28.33
Total (excluding duplicates)	0.02086	0.0465	0.0002613	0.06762	28.33

Clock: clk
Clock Period: 0.083330 usec
Clock Toggle Rate: 23.8183 Mhz
Clock Static Probability: 0.5002

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)
VDD	1.32	0.1562	0.05339	0.02912	0.2387	100

Total Power

Total Internal Power:	0.15620897	65.4360%
Total Switching Power:	0.05339382	22.3667%
Total Leakage Power:	0.02911743	12.1973%
Total Power:	0.23872022	

Hierarchy	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)

soc	0.1438	0.04469	0.008869	0.1974	82.68
accel	0.008519	0.0007123	0.02021	0.02945	12.33
flash_bufs_0_	0	0	2.02e-06	2.02e-06	0.000846
flash_bufs_1_	0	0.0001474	1.631e-06	0.000149	0.06241
flash_bufs_2_	0	0	1.631e-06	1.631e-06	0.0006831
flash_bufs_3_	0	0	1.631e-06	1.631e-06	0.0006831
soc/cpu	0.1175	0.03621	0.007699	0.1614	67.6
soc/memory	9.166e-07	2.81e-05	9.764e-06	3.879e-05	0.01625
accel/fft	0	0	0.01416	0.01416	5.933
accel/mem	0	0	0.005443	0.005443	2.28
soc/spimemio	0.01389	0.003881	0.0005528	0.01833	7.677
soc/simpleuart	0.0107	0.0007944	0.0005005	0.012	5.025
soc/spimemio/xfer	0.0053	0.002085	0.0001595	0.007545	3.16
soc/cpu/genblk1.pcpi_mul	0.0123	4.413e-05	0.0006467	0.01299	5.44
soc/cpu/genblk2.pcpi_div	0.009755	1.654e-05	0.0007201	0.01049	4.395

```
-----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*               Highest Average Power: soc/cpu/CTS_ccl_a_buf_00026 (CLKBUFX20):         0.001994
*               Highest Leakage Power: accel/fft/FE_OFC5361_n_2810 (CLKBUFX20):        2.292e-06
*         Total Cap:      2.99979e-10 F
*         Total instances in design: 73849
*         Total instances in design with no power:     0
*         Total instances in design with no activity:     0
*         Total Fillers and Decap:     0
-----------------------------------------------------------------------------------------
```

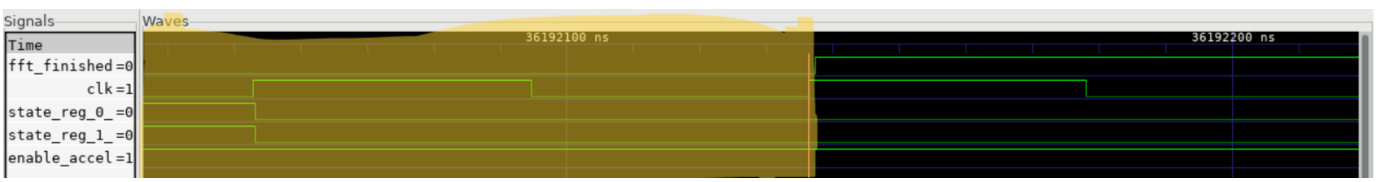


Fig. 12: Zoomed-in waveform illustrating the finished state, where `fft_finished` transitions from 0 to 1

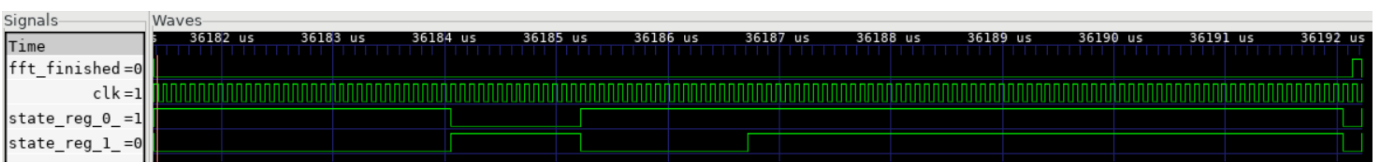


Fig. 13: Complete runtime waveform, starting from the data loading stage(01) and ending at the finish stage(11) when `fft_finished` is asserted (set to 1)

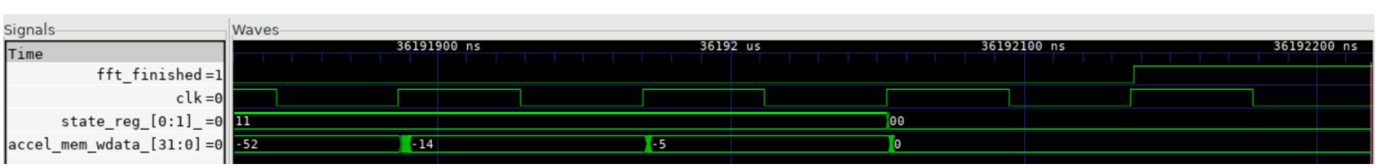


Fig. 14: Waveform showing the final output, which matches the expected value $-14 - 5j$. Where the real and imaginary part are writing back to back.

B.9 Sign-off Simulation Waveforms: